

① Let G be the set of all non-zero real numbers ($\in \mathbb{R} - \{0\}$) and let $a * b = \frac{ab}{2}$ then show that $(G, *)$ is a abelian group.

Proof:

Let $G = \{ a * b = \frac{ab}{2} \text{ for all } a, b \in G \}$

i) Closure axiom:

$$\text{for } a, b \in G \Rightarrow a * b \in G \\ = \frac{ab}{2} \in G$$

$\therefore$ Closure is true.

ii) Associative axiom:

for $a, b, c \in G$

$$a * (b * c) = (a * b) * c$$

$$a * \frac{bc}{2} = \frac{ab}{2} * c$$

$$\frac{abc}{4} = \frac{abc}{4}$$

$\therefore$ Associative is true.

(ii) Identity axiom

for $a \in G$ $\exists$ an identity element 'e' such that

$$a * e = e * a = a$$

$$a * e = a$$

$$\frac{ae}{2} = a$$

$$e = 2 \in G$$

$\therefore$ identity is true.

(iv) Inverse axiom:

for $a \in G$ $\exists$ an inverse element a^{-1} such that

$$a * a^{-1} = a^{-1} * a = e$$

$$a * a^{-1} = e$$

$$\frac{aa^{-1}}{2} = e$$

$$a^{-1} = 4/a \in G$$

$\therefore$ Inverse is true.

$\therefore (G, *)$ is a Group.

(v) Commutative axiom:

for $a, b \in G$ such that

$$a * b = b * a$$

$$\frac{ab}{2} = \frac{ab}{2}$$

$\therefore (G, *)$ is an abelian group.

② Show that $(Z, *)$ is an infinite abelian group where $*$ is defined as $a * b = a + b + 2$.

Sol: (i) Closure axiom:
Since a, b and 2 are integers,
 $a+b+2$ is also an integer.

$$\therefore a+b \in \mathbb{Z}, \forall a, b \in \mathbb{Z}$$

Thus closure axiom is true.

(ii) Associative axiom:

$$\text{Let } a, b, c \in G$$

$$(a+b) * c = (a+b+2) * c$$

$$= a+b+2+c+2$$

$$= a+b+c+4$$

$$a * (b+c) = a * (b+c+2)$$

$$= a + (b+c+2) + 2$$

$$= a+b+c+4$$

$$\Rightarrow (a+b) * c = a * (b+c)$$

Thus associative axiom is true.

(iii) Identity axiom:

Let e be the identity element.

By the definition of e , $a * e = a$

By the definition of $*$, $a * e = a + e + 2$

$$\Rightarrow a + e + 2 = a$$

$$\Rightarrow e = -2.$$

$-2 \in \mathbb{Z}$. Thus identity axiom is true.

(iv) Inverse axiom:

Let $a \in G$ and a^{-1} be the inverse element of a .

By the definition of a^{-1} , $a * a^{-1} = e = -2$

By the definition of $*$, $a * a^{-1} = a + a^{-1} + 2$

$$\Rightarrow a + a^{-1} + 2 = -2$$

$$\Rightarrow a^{-1} = -a - 4$$

Clearly $-a - 4 \in \mathbb{Z}$.

$\therefore$ Inverse axiom is true.

$\therefore (\mathbb{Z}, *)$ is a group.

(v) Commutative property:

Let $a, b \in G$

$$a * b = a + b + 2 = b + a + 2 = b * a$$

$\therefore *$ is Commutative.

$\therefore (\mathbb{Z}, *)$ is an abelian group.

Further, $\mathbb{Z}$ is an infinite set.

The group is an infinite abelian group.

(3) Let G be the set of all rational numbers except 1 and $*$ be defined on G by $a * b = a + b - ab$ for all $a, b \in G$. Show that $(G, *)$ is an infinite abelian group.

Solution:

$$\text{Let } G = \mathbb{Q} - \{1\}$$

Let $a, b \in G$. Then a and b are rational numbers and $a \neq 1, b \neq 1$.

(i) closure axiom:

Clearly $a * b = a + b - ab$ is a rational number. But to prove $a * b \in G$, we have to prove that $a * b \neq 1$.

On the contrary, assume that $a * b = 1$,

$$\text{then } a + b - ab = 1$$

$$\Rightarrow b - ab = 1 - a$$

$$\Rightarrow b(1-a) = 1-a$$

$$\Rightarrow b=1 \quad (\because a \neq 1, 1-a \neq 0)$$

This is impossible, because $b \neq 1$.

$\therefore$ our assumption is wrong.

$\therefore a * b \neq 1$ and hence $a * b \in G$.

$\therefore$ closure axiom is true.

(ii) Associative axiom:

$$a * (b * c) = a * (b + c - bc)$$

$$= a + (b + c - bc) - a(b + c - bc)$$

$$= a + b + c - bc - ab - ac + abc$$

$$(a * b) * c = (a + b - ab) * c$$

$$= (a + b - ab) + c - (a + b - ab)c$$

$$= a + b + c - ab - ac - bc + abc$$

$$\therefore a * (b * c) = (a * b) * c, \quad \forall a, b, c \in G.$$

$\therefore$ Associative axiom is true.

(iii) Identity axiom;

Let e be the identity element.

By definition of e , $a * e = a$

By definition of $*$, $a * e = a + e - ae$

$$\Rightarrow a + e - ae = a$$

$$\Rightarrow e(1-a) = 0$$

$$\Rightarrow e = 0 \text{ since } a \neq 1$$

$$e = 0 \in G.$$

$\therefore$ Identity axiom is satisfied.

(iv) Inverse axiom;

Let a^{-1} be the inverse of $a \in G$.

By the definition of inverse, $a * a^{-1} = e = 0$

By the definition of $*$, $a * a^{-1} = a + a^{-1} - aa^{-1}$

$$\Rightarrow a + a^{-1} - aa^{-1} = 0$$

$$\Rightarrow a^{-1}(1-a) = -a$$

$$\Rightarrow a^{-1} = \frac{a}{a-1} \in G \text{ since } a \neq 1$$

$\therefore$ Inverse axiom is satisfied.

$\therefore (G, *)$ is a group.

(v) Commutative axiom:

For any $a, b \in G$,

$$a * b = a + b - ab$$

$$= b + a - ba$$

$$= b * a$$

$\therefore *$ is Commutative in G and hence

$(G, *)$ is an abelian group. Since

G is infinite, $(G, *)$ is an infinite abelian group.

④ Show that the cube roots of unity forms a finite abelian group under multiplication.

Sol:

Let $G = \{1, \omega, \omega^2\}$ where $\omega^3 = 1$.

The Cayley's table is

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

From the table, we see that

(i) all the entries in the table are members of G . So, the closure property is true.

(ii) multiplication is always associative.

(iii) the identity element is 1 and it satisfies the identity axiom.

(iv) The inverse of 1 is 1.

The inverse of ω is ω^2

The inverse of ω^2 is ω

And it satisfies the inverse axiom

also. $\therefore (G, \cdot)$ is a group.

(v) the commutative property is also true.

$\therefore (G, \cdot)$ is an abelian group.

(vi) Since G is a finite set, $(G, \cdot)$ is a finite abelian group.

⑤ prove that the set of all 4th roots of unity forms an abelian group under multiplication.

Sol: we know that the fourth roots of unity are $1, i, -1, -i$.

Let $G = \{1, i, -1, -i\}$.

The Cayley's table is

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

From the table,

- (i) the closure axiom is true.
- (ii) multiplication is always associative in $\mathbb{C}$ and hence in G .
- (iii) the identity element is $1 \in G$ and it satisfies the identity axiom.
- (iv) the inverse of 1 is 1
 i is $-i$
 -1 is -1
 $-i$ is i

Further it satisfies the inverse axiom. hence $(G, \cdot)$ is a group.

- (v) From the table, the commutative property is also true.

$\therefore (G, \cdot)$ is an abelian group.

Distributivity!

Let $(S, *, \oplus)$ be an algebraic system, where $*$ and $\oplus$ are binary operations on S .

For any $a, b, c \in S$,

$$a * (b \oplus c) = a * b \oplus a * c$$

In this case the operation $*$ is said to be distributive over the operation $\oplus$.

For example, the usual multiplication is distributive over addition, since

$$a \times (b + c) = a \times b + a \times c.$$

Cancellation property:

For any $a, b, c \in S$ and

$$a \neq 0,$$

$$a * b = a * c \Rightarrow b = c$$

and

$$b * a = c * a \Rightarrow b = c$$

For example, Cancellation property holds good for any $a, b, c \in \mathbb{Z}$ under addition and multiplication.

Theorems:

① The identity element of a group is unique.

proof:

Let G be a group.

If possible let e_1 and e_2 be identity elements in G .

Treating e_1 as an identity element, we

$$\text{have } e_1 * e_2 = e_2 \longrightarrow \textcircled{1}$$

Treating e_2 as an identity element,

$$\text{we have } e_1 * e_2 = e_1 \longrightarrow \textcircled{2}$$

From (1) & (2), $e_1 = e_2$

$\therefore$ Identity element of a group is unique.

② The inverse of each element of a group is unique.

proof:

Let G be a group and let $a \in G$

If possible, let a_1 and a_2 be two inverses of a .

Treating a_1 as an inverse of ' a ',

$$a * a_1 = a_1 * a = e \rightarrow \textcircled{1}$$

Treating a_2 as an inverse of ' a ',

$$a * a_2 = a_2 * a = e$$

Now from (1) & (2)

$$a * a_1 = a * a_2$$

$$a_1 = a_2$$

$\Rightarrow$ Inverse of an element is unique.

③ State & prove Cancellation laws on groups.

Let G be a group. Then for all $a, b, c \in G$,

$$(i) \quad a * b = a * c \Rightarrow b = c \quad (\text{Left Cancellation law})$$

$$(ii) \quad b * a = c * a \Rightarrow b = c \quad (\text{Right Cancellation law})$$

proof:

$$(i) \quad a * b = a * c$$

$$\Rightarrow \bar{a}^{-1} * (a * b) = \bar{a}^{-1} * (a * c)$$

(premultiply by $\bar{a}^{-1}$ on both sides)

$$\Rightarrow (\bar{a}^{-1} * a) * b = (\bar{a}^{-1} * a) * c$$

$$\Rightarrow e * b = e * c$$

$$\Rightarrow b = c.$$

$$(ii) \quad b * a = c * a$$

$$\Rightarrow (b * a) * \bar{a}^{-1} = (c * a) * \bar{a}^{-1}$$

$$\Rightarrow b * (a * \bar{a}^{-1}) = c * (a * \bar{a}^{-1})$$

$$\Rightarrow b * e = c * e$$

$$\Rightarrow b = c$$

④ State and prove Reversal law for groups.

Let G be a group & $a, b \in G$.

Then $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof:

It is enough to prove $b^{-1} * a^{-1}$ is the inverse of $(a * b)$.

$\therefore$ To prove (i) $(a * b) * (b^{-1} * a^{-1}) = e$

(ii) $(b^{-1} * a^{-1}) * (a * b) = e$

$$\begin{aligned} \text{(i)} \quad (a * b) * (b^{-1} * a^{-1}) &= a * (b * b^{-1}) * a^{-1} \\ &= a * (e) * a^{-1} \\ &= a * a^{-1} = e \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad (b^{-1} * a^{-1}) * (a * b) &= b^{-1} * (a^{-1} * a) * b \\ &= b^{-1} * (e) * b \\ &= b^{-1} * b = e \end{aligned}$$

$\therefore, b^{-1} * a^{-1}$ is the inverse of $a * b$.

$$\text{ie } (a * b)^{-1} = b^{-1} * a^{-1}.$$

- ⑤ Let G be a group and let a and b be elements of G then
- i) equation $ax = b$ has a unique solution in G & ii) $ya = b$ has a unique solution in G .

proof:

(i) let $x = a^{-1}b$ be the solution of $ax = b$.

$$\therefore a(a^{-1}b) = (aa^{-1})b = eb = b$$

Suppose x_1 & x_2 are two solutions
of $ax=b$ then $ax_1=b$ & $ax_2=b$

$$\Rightarrow ax_1 = ax_2 \quad (\text{by LCL})$$

$$\Rightarrow x_1 = x_2$$

iii) by

(ii) will be proved.

(b) P.T. $(a*b)^2 = a^2 * b^2$ for all
 $a, b \in G$ where G is a group iff
 G is abelian.

Proof:

Let G is abelian

$$\therefore a*b = b*a$$

LHS:

$$\begin{aligned}(a * b)^2 &= (a * b) * (a * b) \\&= a * (b * a) * b \\&= a * (a * b) * b \\&= (a * a) * (b * b) \\&= a^2 * b^2 = \text{RHS}.\end{aligned}$$

Conversely,

$$\begin{aligned}\text{Let } (a * b)^2 &= a^2 * b^2 \\(a * b) * (a * b) &= (a * a) * (b * b) \\a * (b * a) * b &= a * (a * b) * b\end{aligned}$$

$$b * a = a * b \quad (\text{By LCL \& RCL})$$

$\therefore G$ is abelian.

⑦ If $(G, *)$ is an abelian group

then show that $(a * b)^n = a^n * b^n$

for all $a, b \in G$ where n is a +ve integer.

proof:

Let $(G, *)$ be an abelian group.

$$\therefore (a * b) = (b * a) \rightarrow \textcircled{1}$$

For $a, b \in G$, we have

$$(a * b)^1 = (b * a)^1 \quad \text{by (1)}$$

$$\text{and } (a * b)^2 = (a * b) * (a * b)$$

$$= a * (b * a) * b \quad \text{by associativity}$$

$$= a * (a * b) * b \quad \text{by (1)}$$

$$= (a * a) * (b * b) \quad \text{by associativity}$$

$$= a^2 * b^2$$

Thus, the required result is true for $n=1,2$. Let us assume that the result is valid for $n=m$.

$$\text{i.e. } (a * b)^m = a^m * b^m \rightarrow (2)$$

$$\text{Now } (a * b)^{m+1} = (a * b)^m * (a * b)$$

$$= (a^m * b^m) * (a * b)$$

$$= a^m * (b^m * a) * b$$

by associativity

$$= a^m * (a * b^m) * b, \text{ since}$$

G is abelian

$$= (a^m * a) * (b^m * b) \text{ by associativity}$$

$$= a^{m+1} * b^{m+1}$$

Hence, by induction, the result is true for positive integral values of n .

⑧ Show that G is abelian

$$\text{iff } (a * b)^{-1} = a^{-1} * b^{-1}.$$

Proof:

Let G be an abelian.

$$\therefore a * b = b * a \rightarrow \textcircled{1}$$

$$\text{LHS: } (a * b)^{-1} = (b * a)^{-1} \quad (\text{by } \textcircled{1})$$

$$= a^{-1} * b^{-1} \quad (\text{by reversal law})$$

$$= \text{RHS.}$$

Conversely,

$$\text{Let } (a * b)^{-1} = a^{-1} * b^{-1}$$

$$\left((a * b)^{-1} \right)^{-1} = \left(a^{-1} * b^{-1} \right)^{-1} \quad (\text{Take inverse on both sides})$$

$$a * b = (b^{-1})^{-1} * (a^{-1})^{-1}$$

$$a * b = b * a$$

$\therefore G$ is abelian.